

Diffusion Exercise: One-Dimensional Sample Generation

This report shows the final plots from the one-dimensional diffusion model.

The target data is a mixture of two Gaussians with peaks near -4 and 4 . The model is a small MLP. It takes the noisy sample and the time step as input, and it predicts the noise that was added. To generate new samples, I start from Gaussian noise and run the learned reverse diffusion process.

Training Curve

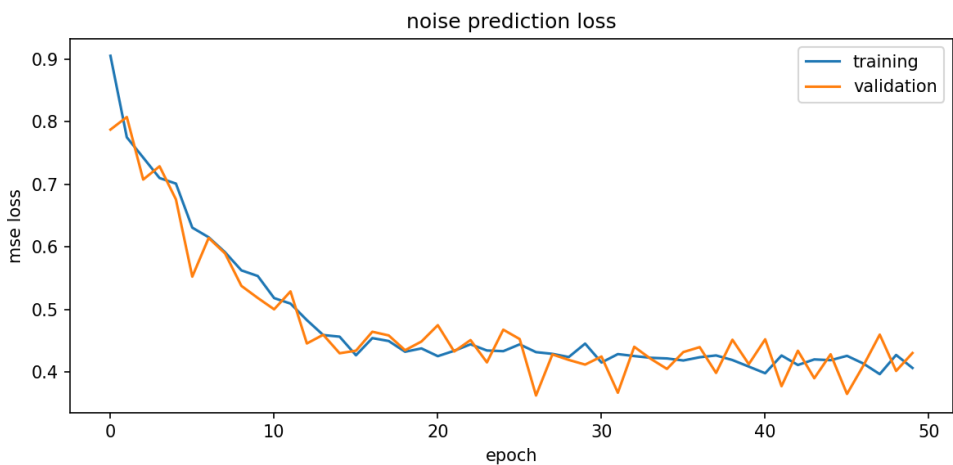


Figure 1: Training and validation loss for predicting the added noise. The loss decreases quickly and then stays roughly constant, so 50 epochs is enough for this small example.

Sampled Distributions During Training

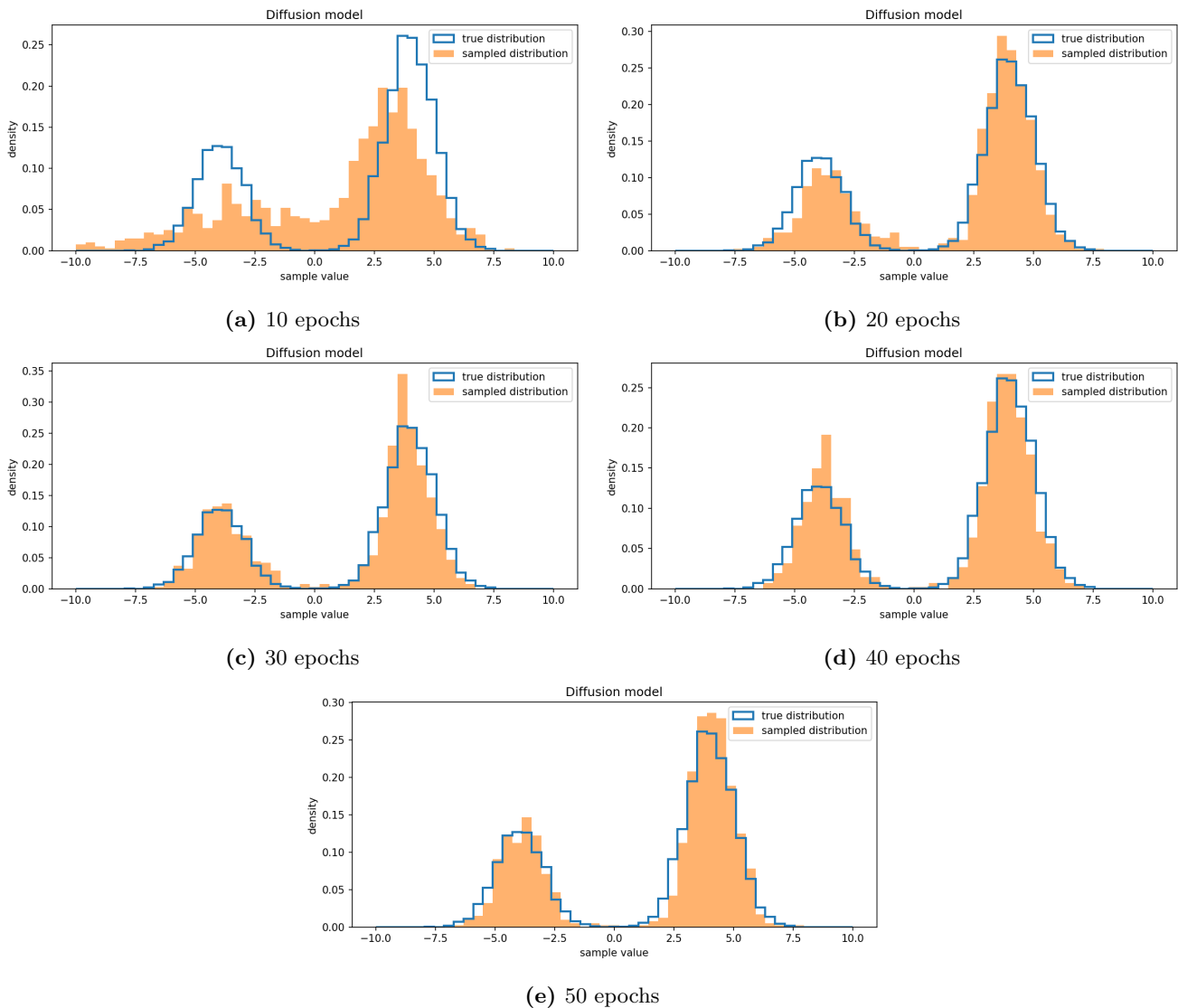


Figure 2: Generated samples during training. The plots are ordered left-to-right and top-to-bottom. The model first learns a rough two-peak shape. By 40–50 epochs, the samples are close to the target distribution.

Reverse Diffusion Process

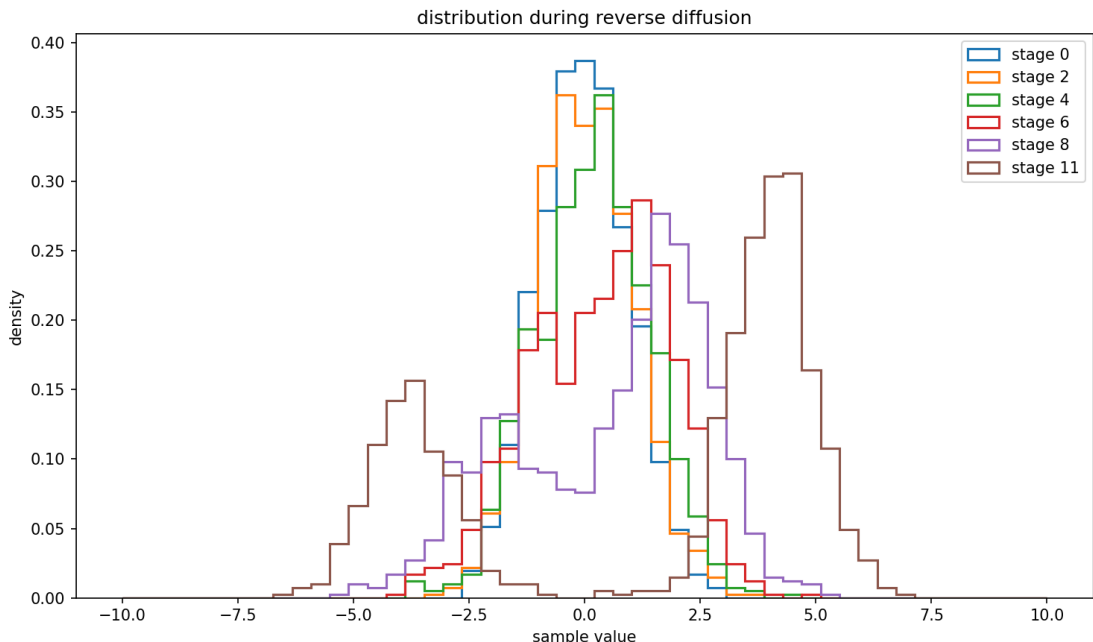


Figure 3: Sample distributions during reverse diffusion. The samples start close to Gaussian noise. During the reverse process, they move into the two modes near -4 and 4 .